Impact of Social Media on Student Life

1. Project Overview and Objective

This Project involves cleaning,transformation,and analyzing raw student Social Mediadatausing Excel and creating an interactive Power BI dashboard to drive meaning insights. The main objective is demonstrates data per-processing techniques using Excel andPowerBI dashboard visualization to understand how social media affects students' sleep, stress, andGPA, and to make informed decisions for students wellness.

2. Data Sources:

- **Source Description : Student Social Media Dataset from Kaggle. • Domain : Education / Psychology / Social Media Analytics** etc

3. Problem Statement :

- To analyse social media usage patterns across Age, Gender, Academic Level to identify high-usage groups.
- To study the relationship between Daily Usage Hours and sleep Duration, sleepQuality and Mental Health Index.
- To evaluate the impact of Late-Night Usage and Social Comparison on Perceived Stress Score.
- To understand how Primary Platform (TikTok, Instagram, YouTube, LinkedIn, Snapchat, X(Twitter), Reddit) Influences Academic Performance (GPA) and Overall Impact.

4. Attribute (Column /Features) Details:

Attribut	Data	
STudent_ID	Text	
Age	Numeric	
Gender	Categorical	
Academic_level	Categorical	
Primary_Platform	Categorical	

Daily_Usage_Hours	Numeric	Average daily social media usage
Weekend_Extra_Hours	Numeric	Average Extra hours usage inWeekend
Device Typ	Text	Smart Phone / Laptop / PC/ Ta
Sleep_Duration_Hours	Numeric	Average sleep hours
Sleep_Quality_Score	Numeric	Average sleep quality s
Late_Night_Usage	Categorical	True / False
Perceived _Stress	Numeric	Average stress score
Mental_Health_Stress	Numeric	Mental Health score(1-10)
Academic_Performance_GPA	Numeric	GPA Score
Overall_Impact	Categorical	Positive / Negative

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:**
- Removed duplicates
- Handled missing value:

Using IF(OR(ISBLANK(M2),M2=0),AVERAGE(M2:M4501),M2) formula tofill theblank value of Preceived_stress_score

Using IF(OR(ISBLANK(O2),O2=0),AVERAGE(O2:O4501),O2)f formula tofill theblank value of Academic_performance_GPA

- **standardized formats, and created calculated fields.**

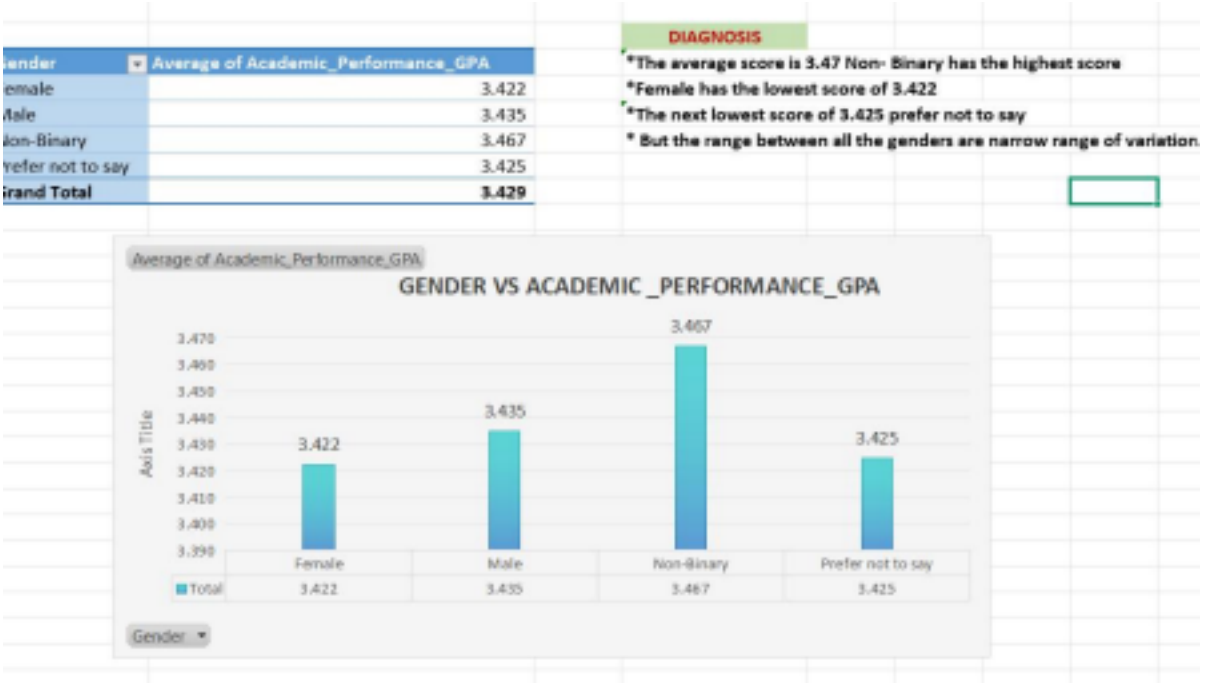
1. =IF(M3>20,"HIGHSTRESS",IF(M3>10,"MEDIUMSTRESS","LOWSTRESS"))2. =IF(O2<2,"LOW GPA",IF(O2<3,"MEDIUM GPA ","HIGH GPA")) • Filtering & Sorting: Organized data to focus on relevant records.



Social_Comparison_Frequency	Perceived_Stress_Score	Mental_Health_Index	Academic_Performance_GPA	Overall_Impact	Stress_Level	GPA_PERFORMANCE
Always	40.00	38.00	1.99	Negative	HIGH STRESS	LOW GPA
Rarely	34.00	36.00	1.97	Negative	HIGH STRESS	LOW GPA
Sometimes	36.00	38.00	1.97	Negative	HIGH STRESS	LOW GPA
Never	35.00	39.00	1.90	Negative	HIGH STRESS	LOW GPA
Sometimes	36.00	43.00	1.95	Negative	HIGH STRESS	LOW GPA
Sometimes	35.00	45.00	1.99	Negative	HIGH STRESS	LOW GPA

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Never	35.00	39.00	1.90	Negative	HIGH STRESS
Sometimes	36.00	43.00	1.95	Negative	HIGH STRESS
Sometimes	35.00	45.00	1.99	Negative	HIGH STRESS

• **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.



Gender	Average of Academic_Performance_GPA
Female	3.422
Male	3.435
Non-Binary	3.467
Prefer not to say	3.425
Grand Total	3.429

DIAGNOSIS

*The average score is 3.47 Non- Binary has the highest score
*Female has the lowest score of 3.422
*The next lowest score of 3.425 prefer not to say
* But the range between all the genders are narrow range of variation



• Convert the data into Fact and Dimension Table **(If Possible)**

=VLOOKUP('Social_media_impact_on_life (1)'!A2,A:D,4,FALSE)

=VLOOKUP('Social_media_impact_on_life (1)'!A2,A:D,2,FALSE)

=VLOOKUP('Social_media_impact_on_life (1)'!A2,A:D,3,FALSE) Using vlookup to create fact and dimension table.

FactTable: Student_ID

Academic_Performance_GPA

Perceived_Stress_Score

Mental_Health_Index

Sleep_Duration

DIM_GPA:

Academic_Performance_GPA

GPA Level

DIM_Stress_level

Perceived_Stress_Score

Stress level

DIM_Students:

Students_ID

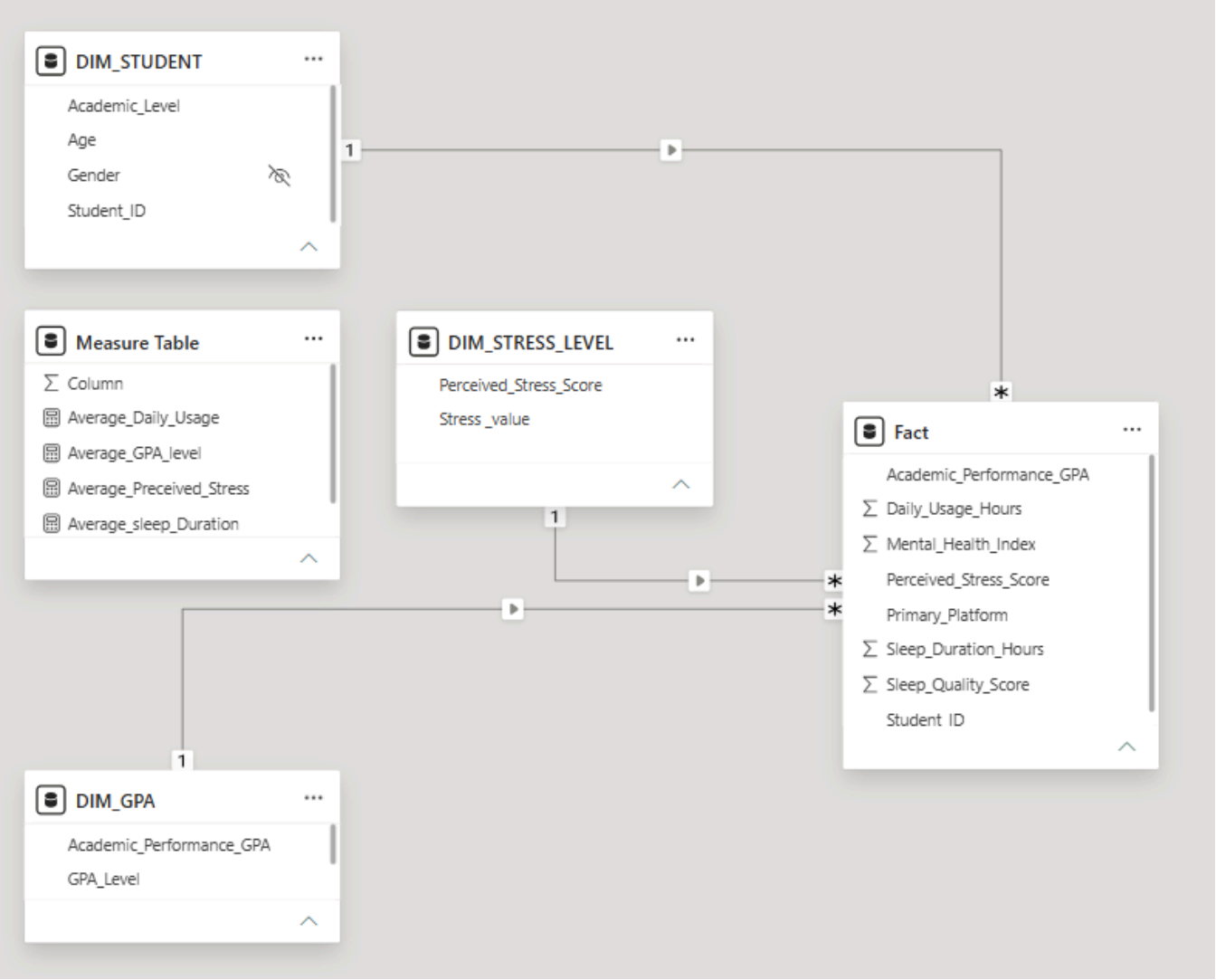
Age

Gender

Academic Level

7. Data Modelling and DAX (Power BI)

• **Data Model:** Established relationships between tables, defined cardinality, andcreated lookup tables where necessary.



- **Calculated Columns & DAX Measures:**

```
Average_Daily_Usage = AVERAGE('Fact'[Daily_Usage_Hours])
Average_GPA_level = AVERAGE('Fact'[Academic_Performance_GPA])
Average_Preceived_Stress = AVERAGE('Fact'[Perceived_Stress_Score])
Average_sleep_Duration = AVERAGE('Fact'[Sleep_Duration_Hours])
High_Stress_count = CALCULATE(COUNTROWS('Fact'),'Fact'[Mental_Health_Index]>=8)
Average_Mental_Health = AVERAGE('Fact'[Mental_Health_Index])
Total_Students = COUNTROWS('Fact')
```

8. Analysis and Visualizations (Power BI)

Dashboard Features:

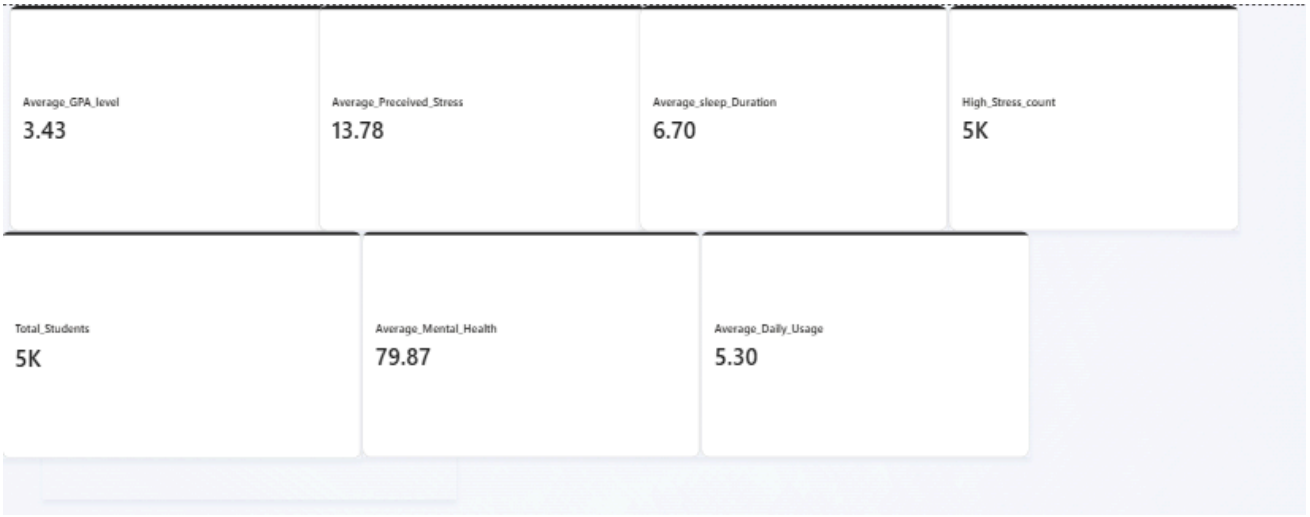
- **Multiple Visualizations based on problem statement:**

KPI cards Total students =5k,

Average GPA Level = 3.43

Average Daily Usage =5.30

Average perceived stress =13.78



BAR CHART - STUDENTS BY PLATFORM

*** Instagram has 1400 students.**

***Tiktok has 1300 and youtube has 1000 .Snapchat, Reddit, X(Twitter) and LinkedIn have fewer students**

COLUMN CHART -AVERAGE OF MENTAL HEALTH_INDEX BY PRIMARY PLATFORM

***The Mental Health index is between 75-80 for all platforms.**

***X(Twitter) is slightly higher and snapchat is lower.**

LINE CHART -Average sleep Duration by Age

***Sleep is stable at 6.6 hours from Age 16 to 22.**

***it drops to 6.4 at age 22-23,**

*** it increases to 7.0 hours at age 24.**

BAR CHART - Average sleep by Platform

***Twitter has the highest sleep at 6.8 hours .**

***Reddit has lowest at 6.5 hours.**

*** Other platforms are between 6.6 to 6.8 hours.**

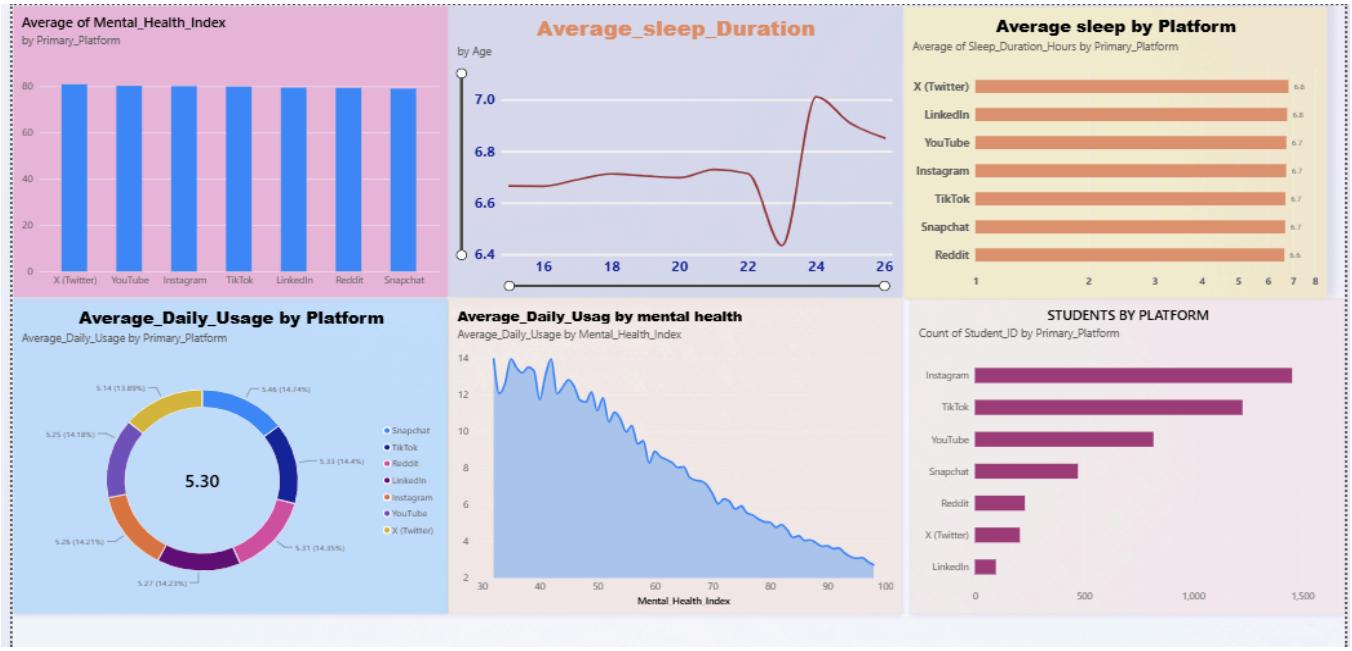
DONUT CHART - Average daily usage by platforms

***Total Average daily usage is 5.30 hours .**

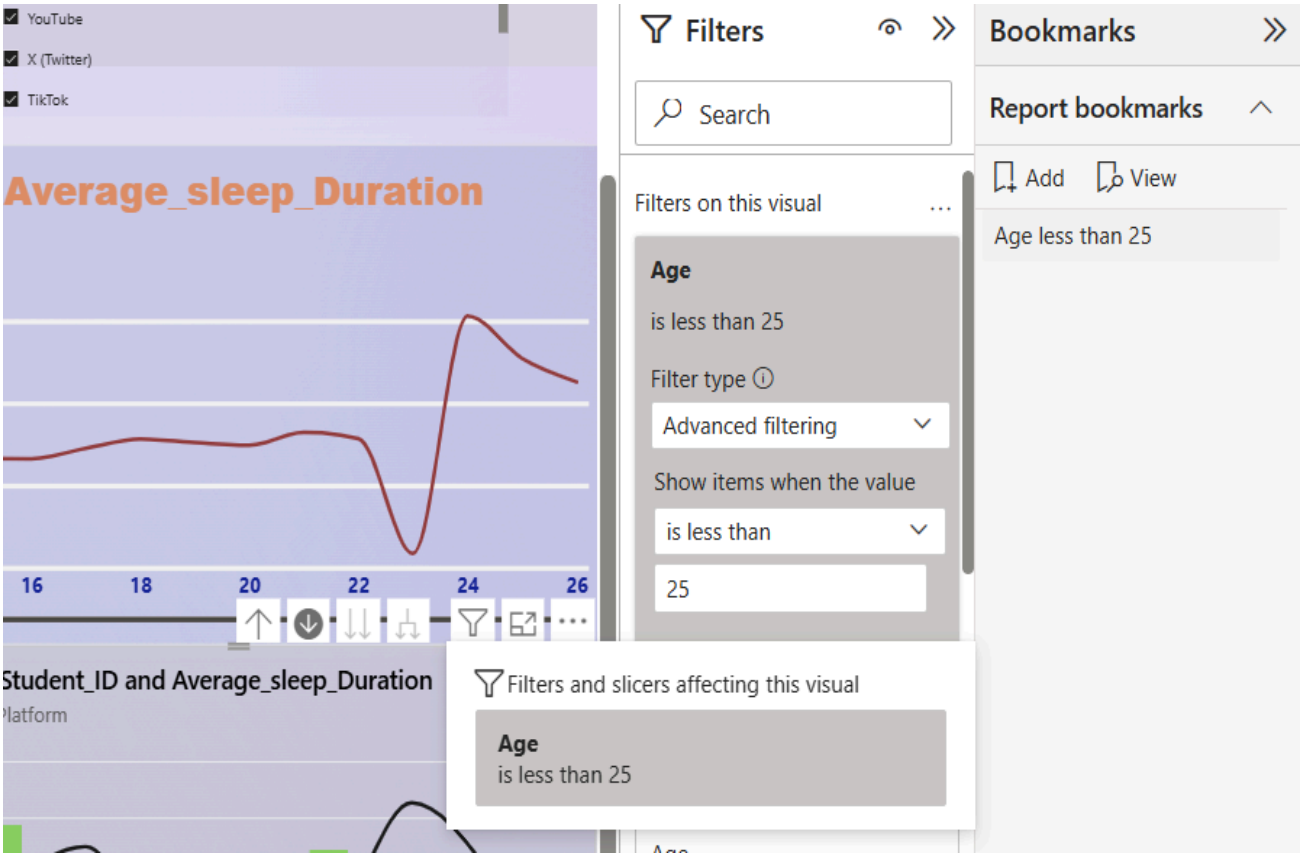
***All Platforms share almost Equal Percentage of around 14% each.**

AREA CHART - Average daily usage by Mental health

- **When Mental health is above 89, usage decreases to 3 hours .**



• Make your Report interactive Layout using Drill-down, Filters and slicers • Bookmarks,



Key Influencer - Average Daily Usage vs Mental Health

*** Daily usage is the main influencer for mental health**

*** When daily usage increases, mental health decreases.**

***Students who use more than 8 hours have very low mental health.**

Sleep Duration -

***When sleep is less than 6hours mental health index is low .**

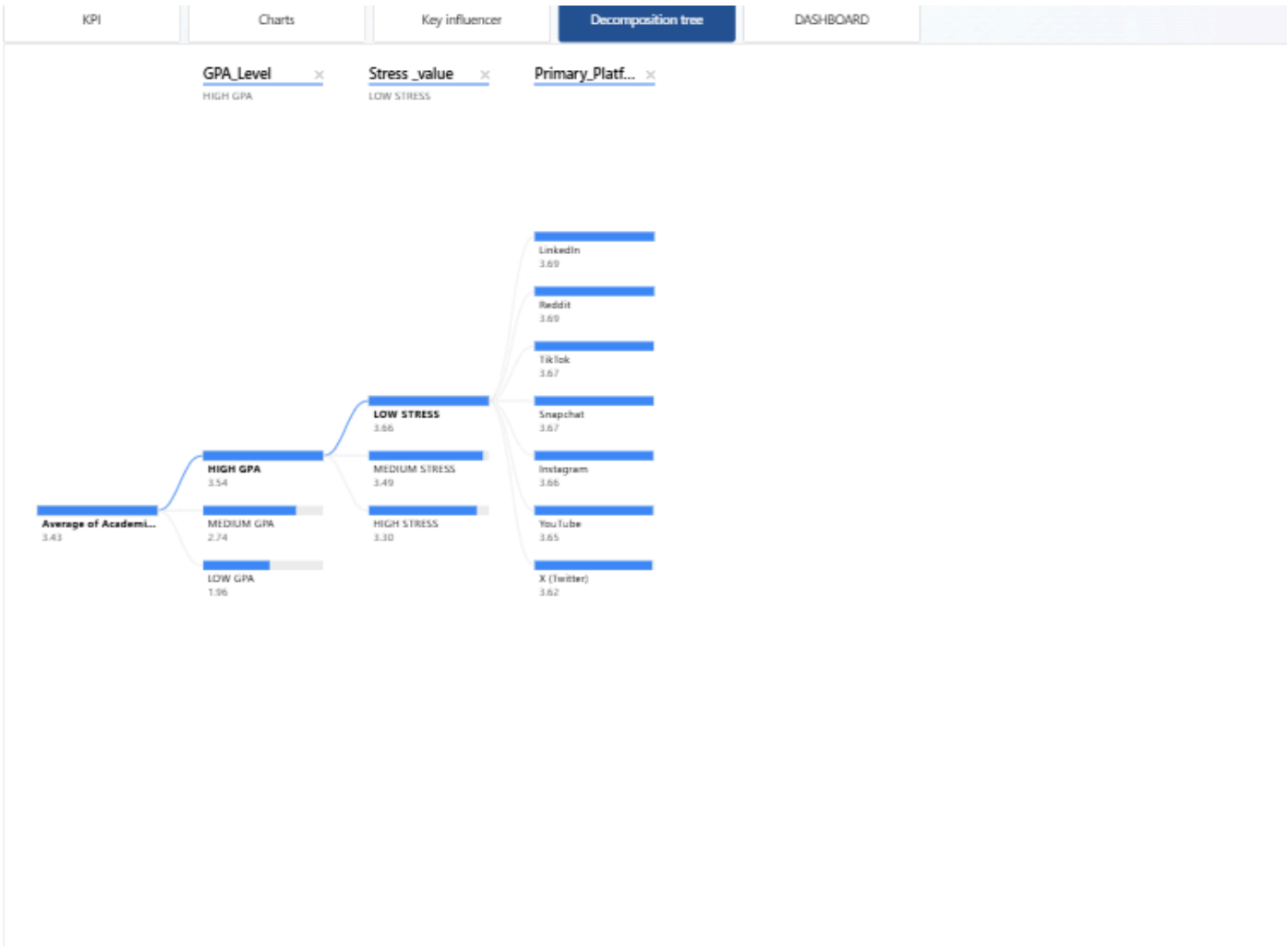
***When the sleep for more than 7 hours, mental health is high.**

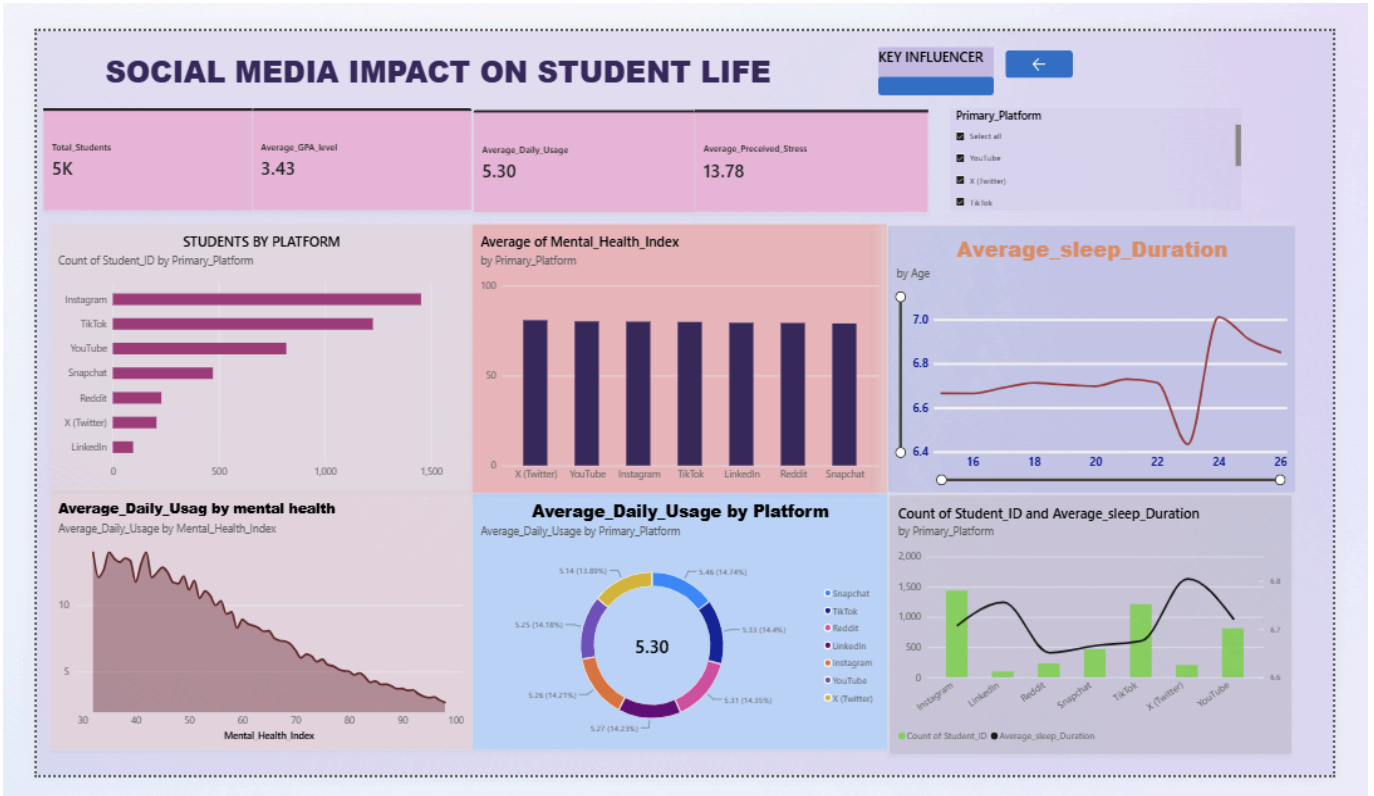
Decomposition Tree - Average of Academic Performance by GPA Level, Stress and Platform

***Total Average is 3.43 first decomposite by GPA Level - High GPA is 3.54 ,Medium GPA is 2.74**

***And Low GPA is 1.96. Students have HIGH GPA decomposite into low stress 3.66. , medium and high .**

***Low stress decomposed by Platform LInkedIn and Raddit are highest at 3.69. Twitter is lowest one.**


KPI
Charts
Key influencer
Decomposition tree
DASHBOARD
GPA_Level
HIGH GPA
Stress _value
LOW STRESS
Primary_Platf...
Average of Academi...
3.43
HIGH GPA
3.54
MEDIUM GPA
2.74
LOW GPA
1.96
LOW STRESS
3.66
MEDIUM STRESS
3.49
HIGH STRESS
3.30
LinkedIn
3.69
Reddit
3.69
TikTok
3.67
Snapchat
3.67
Instagram
3.66
YouTube
3.65
X (Twitter)
3.62


SOCIAL MEDIA IMPACT ON STUDENT LIFE
KEY INFLUENCER
Total_Students
5K
Average_GPA_level
3.43
Average_Daily_Usage
5.30
Average_Preceived_Stress
13.78
Primary_Platform
STUDENTS BY PLATFORM
Count of Student_ID by Primary_Platform
Instagram
TikTok
YouTube
Snapchat
Reddit
X (Twitter)
LinkedIn
Average of Mental_Health_Index
by Primary_Platform
Average_sleep_Duration
by Age
Average_Daily_Usag by mental health
Average_Daily_Usage by Mental_Health_Index
Mental_Health_Index
Average_Daily_Usage by Platform
Average_Daily_Usage by Primary_Platform
5.30
Count of Student_ID and Average_sleep_Duration
by Primary_Platform

9. Insights & Conclusions

- **Key Findings:** • **Provide the analysis insights:**

  ▪ **Descriptive : Total 5000 students were analysed. Average Daily usage is 5.30 hours . Average mental health index is 79.87. Average GPA is 3.43 and their stress level is 13.78. Instagram has the highest number of students with 1300 followed by tik tok and youtube.**

  ▪ **Diagnosis : The reason is daily usage. When usage increases, sleep duration decreased and concentration in studies decreases. The Platform is not the problem , the total time spent is the platform. Students aged 22- 24 sleep the least, only 6.4 hours**

  ▪ **Predictive : if a student uses social media for more than 8 hours, their mental health will fall below 3.0. But if usage is reduced to half of the hours, mental health will go above 85 and GPA will go above 3.6 . Stress is now at 13.78, if it continues like this, it will go above 16 in the next semester.**

  ▪ **Prescriptive: Reduce daily usage to 3-4 hours .**

  ***Keep the phone away 1 hour before sleeping.**

  *** Reduce some hours from instagram and Tiktok and ask them to use Youtube and LinkedIn for Learning .**

  *** So we should conduct awareness programs and provide counseling for High stress students.**

10. Final Conclusions:

From this project, we clearly understand one thing. Social media is not bad, but using it excessively is bad. From the data of 5000 students, we found that students who keep their daily usage low have better mental health, better sleep and better GPA. Reduce some hours from instagram and Tiktok and ask them to use Youtube and LinkedIn for Learning .Reducing total time is the only solution. Finally, we should control social media, social media should not control us. Social media is a great power. If used excessively, it becomes a big problem.